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Louis Ledoux

Associate Professor – Maître de conférences

"Post Hoc Ergo Propter Hoc"

Research Interests

- **Hardware Design & Synthesis:** Floating-Point Units, Systolic Arrays, Tensor Accelerators, FPGAs, ASIC Design, RTL generation
- **Computer Arithmetic:** Number Representations, FloPoCo, Posit & IEEE 754 Standards, Fixed-Point Polynomial Approximations, Accuracy–Energy Tradeoffs
- **High Performance Computing:** BLAS, GEMMs, Heterogeneous Workloads, Numerical Analysis, Supercomputing
- **End-to-End EDA Flows:** From mathematical problem to silicon implementation, integrating MLIR-based compilation, RTL generation, and open-source silicon flows

Experience

2026 – Now **Associate Professor (*Maître de conférences*)**, *Université de Rennes – ISTIC*, Rennes

Teaching-and-research faculty member (*enseignant-chercheur*) at ISTIC, with research in IRISA's Architecture Department (D3) as a member of the Inria Taran project-team.

- Contributing to the design and tapeout of AI accelerators within the national PEPR CAMELIA program.
- Designing systolic-array and tensor architectures with small-format floating-point arithmetic.

2025 – 2026 **Postdoctoral Researcher**, *INSA/INRIA – Emeraude*, Lyon

Worked on arithmetic compilation and optimization within the *Emeraude* team, at the intersection of *MLIR*, *HAriCo/FloPoCo*, *CIRCT*, *Faust*, and open EDA/hardware flows. The research focused on bridging high-level mathematical reasoning with low-level hardware synthesis and reproducible compiler-to-silicon validation.

- Developed multi-level arithmetic transformations and MLIR dialects for fixed-point approximation, ASIC, and FPGA targets.
- Automated hardware generation from DSP code and contributed to FloPoCo's refactoring toward explicit IR-based circuit generation.
- Presented results at *DSD 2025*, *EuroLLVM 2025*, the *PEPR IA Embarquée* workshop in Aussois (2026), and *EuroLLVM 2026*.

2018 – 2024 **Researcher**, *Barcelona Supercomputing Center (BSC) - RoMoL/CAOS/SONAR*, Barcelona

As a BSC researcher during my PhD, I published peer-reviewed papers, attended international conferences, and developed software toolchains and hardware designs for numerical acceleration. The conducted research is summarized as follows:

- Explored co-designed hardware/software acceleration of posit arithmetic for FPGA and ASIC technologies, and POWER9 hosts.
- Developed Kulisch/Quire accumulators for any floating-point representation.
- Conducted accuracy and energy budgeting tailored to workload numerical requirements.
- Designed Systolic Array architecture for HPC workloads with three directions of data flow, infinite number representations, and tailored internal precision.
- Proposed serialized, area-efficient division units targeting the SIMD/Vector paradigm at the architecture level, further improving the parallelism/latency/energy ratio.

2017 - 2018 **Hardware Engineer**, *b⟨⟩com*, Rennes

Coop student under a professionalization contract during my final year of study. Engaged in one year of R&D focused on FPGA acceleration in the cloud, aiming to evaluate the feasibility of these novel solutions.

- Successfully integrated an IP (real-time SDR to HDR) to convert video from NVMe to NVMe through cloud FPGA.
- Developed the IP integration using HDLs and tweaked PCIe drivers (EDMA and XDMA).
- Scheduled user processes asynchronously with OpenCL to overlap reads and writes, successfully saturating PCIe bandwidth (≈15.8 GB/s).
- Traveled to Xilinx XDF Frankfurt 2018 to conduct operational monitoring and analysis.

Keywords: SDAccel, OpenCL, Xilinx FPGA (UltraScale VU9P), VHDL, SystemVerilog, AWS F1 instances, Linux driver, Linux kernel, PCIe, C++1x, C, DMA, AaaS (Acceleration as a Service), FaaS (FPGA as a Service), Virtual Machine, Docker, PCIe virtual functions.

July 2017 **Backend Developer**, *WaryMe*, Rennes
Summer internship focused on developing the entire back end of a people security application.
- Designed and implemented backend services and APIs.
- Ensured secure data transmission using HTTPS and Let's Encrypt.
- Deployed and managed the application on AWS with PM2 and Nginx.
- Automated deployment processes with Jenkins.
- Collaborated with front-end developers using Angular and TypeScript.

Keywords: Node.js, Git, C++, C, Angular, TypeScript, HTTPS, Let's Encrypt, PM2, Nginx, AWS, Jenkins.

July 2016 **Backend Developer**, *ASKIA*, Paris 10ème
During this summer internship, I developed an automated CLI tool for publishing surveys on popular platforms such as GitHub and Zendesk. **Key Concepts:** Node.js, HTTP(S), REST API, Git, Test-Driven Development (TDD), Event-Driven/Asynchronous Programming, Jasmine Framework, Mocks, Stubs.

July 2014 **Electronics Technician**, *Radio Electronique Rennaise (R.E.R)*, Rennes
As an electronics technician, I was responsible for repairing various electronic devices from customers, with an emphasis on audio equipment. Tasks included soldering, reverse engineering amplifier circuits, welcoming customers, and sorting/ordering electronic parts.

Education

2018 – 2024 **PhD Student**, *Universitat Politècnica de Catalunya (UPC)*, Barcelona
Supervisor: Marc Casas Guix. Thesis title: "Floating-Point Arithmetic Paradigms for High-Performance Computing: Software Algorithms and Hardware Designs". Jury members attributed the **excellent** grade. • HAL:tel-04754167.

2015 – 2018 **Engineering School**, *ESIR*, Université de Rennes
Completed a three-year program culminating in an **Engineer diploma** certified by the CTI (Comité des Titres d'Ingénieurs) and a **Master's degree** (Magister) in Computer Science.

2013 – 2015 **Classe Préparatoire**, *CUPGE ESIR*, Université de Rennes
Intensive program with a strong emphasis on Mathematics and Computer Sciences.

Teaching

2026 – 2027 **Course Coordinator – Outils et programmation en C (OPC)**, *Université de Rennes – ISTIC*, Rennes
L3 Computer Science, Informatique and Défi tracks. Responsible for lectures (CM), tutorials (TD), practical sessions (TP), course materials, exam design and marking, and coordination of disability-related study and exam accommodations (PAEH). Linux and shell scripting; C programming, pointers, dynamic memory, data structures, compilation and debugging with GCC, Make, GDB and Valgrind. Programme • Slides and materials • C examples.

2026 – 2027 **Course Coordinator – Unix and Bash refresher (NUnix)**, *Université de Rennes – ISTIC*, Rennes
L3 MIAGE (Computer Science Applied to Business Management), scientific refresher module. Responsible for lectures (CM), tutorials (TD), practical sessions (TP), course materials, exam design and marking, and coordination of disability-related study and exam accommodations (PAEH). Unix/Linux files, permissions, processes, pipes, redirections and Bash scripting. Lectures in English; TD/TP materials in French and English. Programme • Slides and materials • TD/TP worksheets.

13 Aug. 2026 **Course Instructor – Digital Design with MLIR and CIRCT**, *ACM Europe School on MLIR 2026*, A Coruña, Spain
Designed and delivered a 1h45 interactive course for an international audience, using quizzes and guided exercises to teach digital-design fundamentals and practical MLIR/CIRCT flows: IR lowering, optimization, equivalence checking, logic synthesis, and SystemVerilog generation. Developed a capstone in which participants lowered E4M3 floating-point multiplication from `arith.mulf` to `hw/comb` circuitry; demonstrated the end-to-end path from MLIR input to a small Sky130 ASIC GDSII layout, alongside a taped-out TinyTapeout GF0P2 design. Programme • Course slides • source and exercises.

2026 **Computer Architecture (ARC)**, *INSA Lyon*, Lyon
Practical sessions in computer architecture for the Telecommunications curriculum, covering instruction execution, pipelining, cache memories, performance evaluation, and RISC-V assembly programming.
Level: 1st year of engineering school (Telecommunications track, 3TC); Teaching load: 24h

2026 **Operating Systems (SYS)**, *INSA Lyon*, Lyon
Core operating system mechanisms, including kernel and system calls, process scheduling, virtual memory, concurrency, and file systems.
Level: 1st year of engineering school / Bachelor equivalent (Computer Science, INSA 3IF); Teaching load: 16h

2026 **Long-Term Project in Compilation (PLD-COMP)**, *INSA Lyon*, Lyon
Supervision of long-term group projects on compiler design and implementation, covering intermediate representations, program analysis, and backend code generation.
Level: 2nd year of engineering school / Master 1 (Computer Science, INSA 4IF); Teaching load: 36h

Teaching (continued)

2025 – 2026 **Computer Architecture (AO)**, *INSA Lyon*, Lyon
Problem-solving and practical sessions on processor organization and instruction set architectures, based on the MSP430 microcontroller, including instruction execution, addressing modes, and memory hierarchy.
Level: 1st year of engineering school / Bachelor equivalent (Computer Science, INSA 3IF); Teaching load: 12h

2025 – 2026 **Digital Design (AC)**, *INSA Lyon*, Lyon
Exercises and guided sessions on digital circuit design, including combinational and sequential logic, finite state machines, and introductory hardware description languages.
Level: 1st year of engineering school / Bachelor equivalent (Computer Science, INSA 3IF); Teaching load: 12h

International Peer-reviewed Journals

[Bar+26a] B. Barbe, **L. Ledoux**, A. Volkova, and F. de Dinechin, "Reconfigurable constant multipliers: hardware models, optimization algorithm and applications," *Microprocessors and Microsystems*, vol. 122, article 105270, June 2026. • DOI • HAL:05510936.

[Pop+25] M. Popoff, R. Michon, T. Risset, P. Cochard, **L. Ledoux**, S. Letz, Y. Orlarey, and F. de Dinechin, "Frugalité et conception de circuits pour le traitement du signal audio numérique," *Revue Francophone d'Informatique et Musique*, special issue "Frugalité, pérennité et création," vol. 11, 2025. • DOI • HAL:05489376.

[Led+25b] **L. Ledoux**, P. Cochard, and F. de Dinechin, "Towards optimized arithmetic circuits with MLIR," *WiPiEC Journal* (Works in Progress in Embedded Computing), vol. 11, no. 1, 2025, Salerno, Italy. Associated conference presentation at the Euromicro Conference on Digital System Design (DSD). • DOI • HAL:05385229.

International Peer-reviewed Conference Papers

[Bar+26b] B. Barbe, **L. Ledoux**, F. de Dinechin, and A. Volkova, "RSCM-Net: FPGA-Aware Co-Design of Quantized DNNs with Reconfigurable Single Constant Multipliers," *International Conference on Field Programmable Technology (FPT 2026)*, Honolulu, Hawaii, USA, December 7–10, 2026. *Forthcoming.* • Conference.

[Led25] **L. Ledoux**, "Design-space exploration of serialized floating-point division for DLP architectures," in *28th Euromicro Conference Series on Digital System Design (DSD 2025)*, Salerno, Italy. • DOI • HAL:05385247.

[LC23a] **L. Ledoux** and M. Casas, "An open-source framework for efficient numerically-tailored computations," in *33rd International Conference on Field-Programmable Logic and Applications (FPL 2023)*, Gothenburg, Sweden. • DOI • arXiv:2406.02579 • HAL:04277512.

[LC22] **L. Ledoux** and M. Casas, "A generator of numerically-tailored and high-throughput accelerators for batched GEMMs," in *30th IEEE International Symposium on Field-Programmable Custom Computing Machines (FCCM 2022)*, May 2022, pp. 1–10, New York, USA. • DOI • HAL:04103774.

Tutorials, Workshops & Demos

MLIRW26 **7th MLIR Workshop @ EuroLLVM 2026**, Dublin, Ireland, April 13, 2026. *Talk.* Progressive Arithmetic Lowering from Tensor Kernels to Synthesizable Datapaths. • HAL:05594483.

ARITH26 *Tutorial: Optimizing Datapath Circuits with MLIR and CIRCT*, with Sam Coward, June 27, 2026. Exercises and slides.

Demo: Printing Mathematical Intent onto Silicon with Emeraude-MLIR, HAriCo, and OpenROAD, June 29, 2026, Fulda, Germany. Programme.

Contributions to Publications

[Ven24] Contributor (not a listed author) to: M. Venn, "Tiny Tapeout: a shared silicon tapeout platform accessible to everyone," *IEEE Solid-State Circuits Magazine*, 2024. Initiated manuscript drafting, ensured academic formatting, and contributed to revision. • IEEE article • manuscript commits.

Poster Presentations

[Led26e] **L. Ledoux**, "HAriCo: Floating-Point MLIR to CIRCT Datapaths," in *ACM Europe School on MLIR 2026*, A Coruña, Spain, August 10–14, 2026. • HAL:05725118.

[Led26d] **L. Ledoux**, "Multi-Level Compilation of AI Tensors and Activation Functions to Silicon," in *PEPR IA Days*, Rennes, France, June 22–24, 2026. • HAL:05671676.

[Led+26c] **L. Ledoux**, P. Cochard, and F. de Dinechin, "Progressive Arithmetic Optimizations: Tensor and DSP Kernels to Synthesizable Datapaths," • *3rd FPGA Developers' Forum (FDF) meeting*, CERN, May 27–29, 2026. • HAL:05631193v1.

[Led+26b] **L. Ledoux**, P. Cochard, and F. de Dinechin, "Floating-Point Datapaths in CIRCT via HAriCo and harico-arith-to-comb Lowering," in *EuroLLVM Developers' Meeting 2026*, Dublin, Ireland. • HAL:05576131.

[Led+26a] **L. Ledoux**, P. Cochard, and F. de Dinechin, "Arithmetic lowering with Emeraude-MLIR: bridging tensor and DSP kernels to silicon datapaths," Aussois, France. • HAL:05489427.

[Led+25a] P. Cochard, L. Forget, F. de Dinechin, and **L. Ledoux**, "Towards multi-level arithmetic optimizations," in *EuroLLVM 2025*, Berlin, Germany. • HAL:05063466.

[LC24b] **L. Ledoux** and M. Casas, "LLMMMM: large language models matrix–matrix multiplications characterization on open silicon," in *11th BSC Severo Ochoa Doctoral Symposium 2024*, Barcelona, Spain. • HAL:04592229.

[LC24a] **L. Ledoux** and M. Casas, "The grafted superset approach: bridging Python to silicon with asynchronous compilation and beyond," in *4th Workshop on Open-Source Design Automation (OSDA 2024)*, hosted at *DATE*, Valencia, Spain. • HAL:04587458.

[LC23b] **L. Ledoux** and M. Casas, "Open-source GEMM hardware kernels generator: toward numerically-tailored computations," in *10th BSC Severo Ochoa Doctoral Symposium 2023*, Barcelona, Spain. • arXiv:2305.18328 • HAL:04094835.

Invited Talks & Seminars

[LC25] P. Cochard and **L. Ledoux**, "FloPoCo and MLIR: a multi-level compilation framework for many intents," Holigrail Seminar, Sorbonne University, Paris, France, September 18, 2025. • HAL:05412130.

[Led24] **L. Ledoux**, "The walls and the dark silicon era: an arithmetic perspective," seminar at Inria Rennes (Team TARAN), Rennes, France, December 2, 2024. • HAL:04823130.

[LC19] **L. Ledoux** and M. Casas, "Accelerating DL inference with (Open)CAPI and posit numbers," in *OpenPOWER Summit Europe 2019*, Lyon, France. • HAL:04094850.

Educational Videos & Scientific Outreach

Zero to ASIC **Arbitrary-precision floating-point systolic arrays.** Educational presentation of a systolic-array generator with configurable floating-point formats and accumulator precision, for mentoring and teaching in the Zero to ASIC and Tiny Tapeout communities. • Educational video (YouTube).

Zero to ASIC **Visualization-oriented global placement.** Educational presentation for Matt Venn's Zero to ASIC course on shaping a chip during global placement while exposing the underlying physical-design process. • Lecture recording (Google Drive).

SUF demo **64×64 systolic-array placement.** Video demonstration of the SUF flow's placement visualization for a 4,096-PE array in ASAP7. • YouTube demonstration.

Moduland **Moduland festival (GRAME–cncm).** Staffed a booth for nine hours on behalf of the Emeraude team (INSA–Inria–GRAME), presenting an electronic instrument and discussing links between sound, numerical computing, and integrated-circuit design. • Instagram.

Conference Peer Reviewing

2026 FCCM 2026 (×1), ASAP 2026 (×2), and ICECS 2026 (×1).

Conference Organization Experience

LAC25 **Linux Audio Conference**, Lyon, France (2025). *Technician.* Responsible for sound, YouTube streaming, and microphone management. event website.

JIM25 **Journées de l'Informatique Musicale**, Lyon, France (2025). *Technician.* Responsible for sound, YouTube streaming, and microphone management. event website.

HC32 **HotChips 32nd**, Remote (Digital), SARS-CoV-2 Period (2020). *Remote Assistant.* Assisted with digital entrance management, conference link distribution, and attendee support during the event. event website.

FPL29 **Field-Programmable Logic Conference 29th**, Barcelona, Spain (2019). *Reception and Entrance Coordinator.* Managed reception desk activities, including badge distribution and providing microphones for speakers during presentations. event website.

Research Project & Internship Mentoring

2026 – 2027 **Thomas Brembilla** (ENS Rennes, M2) and **Edouard Pierre** (ISTIC, M1 Electronics). *Ongoing co-supervision* with Noë Mercier-Brosse and Olivier Sentieys for the 7th Thales RISC-V student competition: acceleration of the QOA audio encoder on the CVA6 RISC-V processor. • 2026–2027 announcement.

2026 – 2027 **ENS Rennes – M1 SIF research project (ongoing)**. "Vector Floating-Point Architectures for Softmax and Exponential Acceleration," co-supervised with Sami Ben Ali and Olivier Sentieys in the TARAN team (Inria/IRISA). Study of vector FPUs and exponential approximations, balancing softmax throughput, hardware cost, and numerical accuracy within CAMELIA. • Project topic (PDF) • ENS project listing.

2026 **Esteban Wybouw**, "Porting FloPoCo to HAriCo: an IR-based core generation framework for multi-HDL hardware generation," co-supervised with Florent de Dinechin and Pierre Cochard. Internship completed in August 2026.

2026 **Azure Blondon, Hugo Berthet-Rambaud, Reda Id Hammouche, and Simon Cabias**. L3 / first year of the engineering cycle (INSA Lyon, 3IF): research internships in the FloPoCo/HAriCo framework. Personal supervision share: approximately 15%, alongside lead supervisors Pierre Cochard and Florent de Dinechin.

Academic Summer Schools and Workshops

2026 **ACM Europe School on MLIR**, A Coruña, Spain, August 10–14. *Instructor.* Delivered "Digital Design with MLIR and CIRCT" (1h45); see Teaching for course materials. Programme.

2021 **ACM Europe Summer School on High Performance Computing**, Barcelona, Spain. *Participant.* Attended lectures and hands-on sessions covering exascale architectures, programming models, and numerical methods. school website.

2019 **Yale Patt's Computer Architecture Summer School**, Barcelona, Spain. *Participant.* Attended the special summer edition of Prof. Yale Patt's course on out-of-order execution and advanced microarchitectural design.

2019 **Xilinx Machine Learning and Artificial Intelligence Seminar**, Paris, France. *Participant.* Technical seminar focused on machine learning workloads and AI acceleration platforms.

2019 **PRACE Summer School on Heterogeneous Programming with OmpSs@FPGA**, Barcelona, Spain. *Participant.* Training on FPGA-based acceleration and heterogeneous programming using the OmpSs@FPGA model.

Technology Development: Software & Hardware

Software SUF: OpenROAD-based ASIC design-space exploration and flow orchestration; OSFNTC: numerically-tailored GEMM generator with PyTorch/NumPy/OpenBLAS/OpenCAPI integration; MLModelMLIRGEN: PyTorch/`torchvision`→MLIR exporter.

Hardware IP POF: SystemVerilog Posit operators, datapaths, quires, and verification benches; FloPoCo: Virtex UltraScale+ timing model, systolic-array generator work, and HAriCo/IR-oriented refactoring.

Utilities VH2V: VHDL→Verilog bridge for open ASIC flows; OpenROAD GPL: placement visualization and chip-shaping experiments; gdsiistl: GDSII→STL layout visualization; Codez: Typst/CeTZ annotated code figures with MLIR/Python/SystemVerilog support.

Upstream CIRCT PR9263 and PR9245: arithmetic lowering and HDL-oriented transforms; oc-accel and capi2-bsp: OpenCAPI/CAPI2 accelerator support; OpenROAD PR: Arch/Manjaro installer support.

Tapeouts MPW5 Sky130: posit/Quire systolic array; TT GF0P2: Faust→MLIR→silicon soft-clipper; TT SKY26A: MLIR→floating-point accumulator; IHP BF16 and IHP IEEE754: CMOS5L floating-point lowering demonstrators; TT SKY25B: modified-placement demonstrator; MPW1 and MPW4: integration contributions.

Languages

French Native

Spanish Native (with an honest French accent)

English Full Proficiency